

Planned numerical sampling and constraint construction for an MKM channel DNS at $Re_\tau = 395$

Prepared for the Shenfun MKM channel-flow workflow

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Abstract

This note specifies a gated plan for a new turbulent-channel direct numerical simulation at nominal friction Reynolds number $Re_\tau = 395$. It adapts the existing two-stage Shenfun MKM workflow to the mesh and domain used by Moser, Kim, and Mansour for their corresponding case. Particular attention is paid to the reference paper’s odd wall-normal count, the distinction between the reference Chebyshev–Gauss–Lobatto convention and Shenfun’s Chebyshev–Gauss quadrature grid, the even-size requirement of the optimized Shenfun biharmonic solver, the constant-flux target, integer de-aliasing sizes, measured wall stress, sampling convergence, constraint construction, and the much larger storage cost of an all-plane temporal record. The proposed persistent representation stores the two independent MKM spectral state variables and the required zero horizontal modes, from which all three velocities on all 192 horizontal planes are reconstructed without approximation beyond solver roundoff. This is a planning document: no $Re_\tau = 395$ production simulation or result is claimed here.

Status: the Ubuntu and CentOS 18-plane and 192-plane numerical/I/O gates are verified. The former full-resolution $\Delta t = 2.5 \times 10^{-4}$ pilot was unstable; the refined-step pilot, physical stationarity, and resolution-acceptance gates remain pending.

1 Objective and reference case

Let $x \in [0, L_x)$ be streamwise, $z \in [0, L_z)$ be spanwise, and $y \in [-h, h]$ be wall-normal. The proposed nondimensional channel has

$$h = 1, \quad L_x = 2\pi h, \quad L_z = \pi h, \quad (1)$$

with periodicity in x and z and no slip at $y = \pm h$. Velocities are ordered as

$$\mathbf{u} = (u_x, u_z, u_y)^T$$

in the target files: streamwise, spanwise, and wall-normal. The internal MKM HDF5 groups remain

$$\mathbf{u0} = u_y, \quad \mathbf{u1} = u_x, \quad \mathbf{u2} = u_z.$$

Moser, Kim, and Mansour reported the parameters in Table 1. Their nominal case name is $Re_\tau = 395$, while the measured value in the paper and statistical database is 392.24.

The source paper states that domain sizes were chosen so two-point correlations were essentially zero at half-box separation and that resolution was chosen so energy spectra were sufficiently small at the largest retained wavenumbers. It also reports at least 13 wall-normal points below $y^+ = 10$. These are validation conditions, not merely historical metadata.

The primary reference data are available from the University of Texas turbulence database:

https://turbulence.odon.utexas.edu/MKM_1999.html.

Table 1: Moser–Kim–Mansour reference parameters for the nominal $Re_\tau = 395$ channel.

Quantity	Reference value
Actual Re_τ	392.24
$(L_x, L_z)/h$	$(2\pi, \pi)$
Paper order $N_x \times N_y \times N_z$	$256 \times 193 \times 192$
Δx^+	approximately 10.0
Δz^+	approximately 6.5
Δy_c^+	approximately 6.5

2 Proposed Shenfun mesh

2.1 Ordering and nominal spacings

The current driver uses solver order

$$(N_{\text{wall}}, N_{\text{stream}}, N_{\text{span}}).$$

The user-selected Shenfun baseline is

$$\boxed{\text{--n } 192 \ 256 \ 192}. \quad (2)$$

At nominal $Re_\tau = 395$, its two uniform-direction spacings are

$$\Delta x^+ = \frac{L_x}{N_{\text{stream}}} \frac{u_\tau}{\nu} = \frac{2\pi(395)}{256} = 9.69476, \quad (3)$$

$$\Delta z^+ = \frac{L_z}{N_{\text{span}}} \frac{u_\tau}{\nu} = \frac{\pi(395)}{192} = 6.46317. \quad (4)$$

The saved wall-normal mesh is Shenfun’s Chebyshev–Gauss quadrature grid,

$$y_m = \cos\left(\frac{(2m+1)\pi}{2N_{\text{wall}}}\right), \quad m = 0, \dots, N_{\text{wall}} - 1. \quad (5)$$

For even $N_{\text{wall}} = 192$, the centerline lies midway between the two central Gauss points. The gap across the center, distance of either central point from the center, and first saved distance from a wall are

$$\Delta y_c^+ = 2(395) \sin\left(\frac{\pi}{2(192)}\right) = 6.46310, \quad (6)$$

$$y_{\text{nearest center}}^+ = 395 \sin\left(\frac{\pi}{2(192)}\right) = 3.23155, \quad (7)$$

$$y_1^+ = 395 \left[1 - \cos\left(\frac{\pi}{2(192)}\right)\right] = 0.01322. \quad (8)$$

There are 14 positive-half-channel Gauss nodes with distance below $y^+ = 10$ from the nearest wall.

2.2 Why the paper uses 193 but Shenfun uses 192

The paper does not identify an odd count as a separate physical requirement. The most direct numerical explanation is the conventional Chebyshev–Gauss–Lobatto representation. If the maximum polynomial degree is even, the number of endpoint-including nodal values is one greater and hence odd. Counts 129, 193, and 257 correspond to even interval/degree counts 128, 192,

and 256; this also includes both walls and the centerline and is convenient for cosine-transform implementations.

Shenfun’s grid in Eq. (5) is not the same nodal family: the wall values are excluded and no slip is imposed through composite Galerkin bases. Although its transforms and bases can represent odd sizes, the optimized Chebyshev biharmonic solver used by the MKM wall-normal-velocity equation splits the coefficient system into equal even/odd parity blocks. A four-rank Ubuntu test at $N_{\text{wall}} = 17$ was finite after initialization but generated non-finite velocity coefficients in the first Runge–Kutta substage. Its matched $N_{\text{wall}} = 18$ control remained finite for 40 steps and reconstructed every saved velocity value exactly from the independent state.

The current solver must therefore use an even wall-normal size. The selected $N_{\text{wall}} = 192$ is one point, or approximately 0.52 percent, below the paper’s count. It retains 14 saved points per wall below $y^+ = 10$ and gives a center gap essentially equal to the reference center spacing.

The two Fourier counts should remain even. This retains efficient FFT sizes, the expected real-transform and Nyquist conventions, and convenient MPI factorizations.

2.3 Integer de-aliasing size

The nonlinear terms use padded transforms. The current implementation replaces a requested padding factor p by an integer size

$$N_p = \lfloor pN \rfloor.$$

A literal $p = 1.5$ at $N = 192$ gives the exact integer padded size 288. The proposed integer padded sizes are

$$(N_{p,y}, N_{p,x}, N_{p,z}) = (288, 384, 288), \quad (9)$$

so the run card should use

$$\text{--padding-factor } 1.5 \ 1.5 \ 1.5. \quad (10)$$

3 Reynolds number and constant-flux target

The nominal nondimensional inputs are

$$h = 1, \quad u_{\tau,\text{ref}} = 1, \quad \nu = \frac{1}{395}. \quad (11)$$

However, prescribing ν does not by itself demonstrate that the developed constant-flux solution has $Re_\tau = 395$. If the time-averaged wall shear is $\bar{\tau}_w$, the measured friction velocity and Reynolds number are

$$u_{\tau,\text{meas}} = \sqrt{\bar{\tau}_w}, \quad Re_{\tau,\text{meas}} = \frac{u_{\tau,\text{meas}} h}{\nu}. \quad (12)$$

Both walls must be evaluated and their symmetry reported.

The current $Re_\tau = 180$ runner defaults to an $Re_\tau = 180$ bulk velocity and must not be reused unchanged. Numerical integration of the official MKM $Re_\tau = 392.24$ mean profile gives the planning value

$$\frac{U_b}{u_\tau} = 17.5447511436. \quad (13)$$

For Eq. (1), the corresponding integrated flux is

$$Q = 2hL_xL_zU_b = 4\pi^2U_b = 692.639012411. \quad (14)$$

This is an initial calibration target. The developed simulation is accepted only if Eq. (12) is within the agreed tolerance; otherwise U_b is adjusted and a calibration continuation is run.

4 Initialization changes required before execution

Changing only `--re` is insufficient. The present `demo/MKM.py` contains an $Re_\tau = 180$ initial centerline/profile constant, the old bulk-velocity default, and perturbation choices whose wall-unit wavelengths change with Reynolds number. Before production, the run metadata must record:

1. nominal Re_τ , ν , target U_b , and integrated flux;
2. the initial mean profile and its source;
3. perturbation amplitudes and wavelengths in wall or outer units;
4. the realized integer padded sizes;
5. a checksum or version identifier for the reference profile.
6. an all-plane state output path for the independent wall-normal velocity and wall-normal vorticity coefficients, the two separately evolved zero-horizontal-mode profiles, online block moments, and infrequent complete audit fields. The current runner exposes only one full-field `modsave` cadence, so this recorder and its decoder must be implemented and smoke-tested before stationary sampling.

The recommended clean start interpolates the official MKM mean profile onto Eq. (5), reflects it across the centerline, and adds small resolved perturbations. The interpolated profile is only an initial condition and validation reference. It is not counted as new DNS evidence. The initialized velocity must pass divergence and no-slip checks after the same forward/backward projections used by the solver.

5 Preflight gates

5.1 Even-grid and restart smoke

First run a small even wall-normal case

$$(N_{\text{wall}}, N_{\text{stream}}, N_{\text{span}}) = (18, 32, 24),$$

through initialization, a forced checkpoint, restart, quadrature HDF5 output, state reconstruction, and a complete NaN/Inf audit. The uninterrupted and restarted states must agree to solver tolerance.

5.2 Full-shape decomposition benchmark

Allocate the unpadded shape (192, 256, 192) and padded shape (288, 384, 288) on the Linux server and take only a few steps without snapshots. Benchmark feasible MPI sizes, initially 16, 32, and 64 ranks. The production rank count is selected from successful decomposition, memory, and step-time measurements.

The isolated Ubuntu pilot completed two full-shape steps at all three rank counts. The 64-rank one-sample I/O gate then reconstructed all 192 planes with maximum absolute component error 2.14×10^{-14} , maximum relative error 1.06×10^{-15} , maximum divergence 9.53×10^{-16} , and no non-finite values. The independent-state and physical-velocity files were 152,594,328 and 226,523,696 bytes, respectively. A reduced spin-up followed by two sampling segments also passed checkpoint restart, HDF5 append, state-shard continuity, and reconstruction checks. CentOS jobs 3201–3204 subsequently passed the 16-, 32-, and 64-rank allocation gates and the 64-rank state-I/O gate. The requested scheduler policy allocates two SLURM tasks per launched MPI rank: a 64-rank run requests `#SBATCH --ntasks=128`, while the launcher explicitly uses `mpiexec -n 64`. The shared script validates and records this 2:1 allocation-to-launch ratio.

5.3 Time-step pilot

The full-resolution CentOS pilot in job 3206 used

$$\Delta t = 2.5 \times 10^{-4}. \quad (15)$$

This step is rejected. The reported CFL exceeded one at $t = 0.1$, approached 1.6, and the spanwise kinetic energy grew from 0.3907 to 12.8004 by $t = 0.8$. At $t = 0.9$, all 28,311,552 physical velocity values were non-finite. The minimum Chebyshev–Gauss wall spacing at $N_{\text{wall}} = 192$ is about one ninth of that at 64 points. Scaling the old $Re_\tau = 180$ step 5×10^{-4} by this ratio gives 5.56×10^{-5} . The conservative candidate used in the revised scripts is therefore

$$\boxed{\Delta t = 5 \times 10^{-5}}. \quad (16)$$

The first full-mesh refined pilot ends at $t = 1$ and records diagnostics every 0.1 outer units. Finite values, CFL margin, energy, divergence, restart consistency, and high-wavenumber spectra determine acceptance.

The refined step and cadence passed a four-rank, 200-step (18, 32, 24) Ubuntu round trip: all 20 retained samples reconstructed exactly, the finite-value audit passed, maximum divergence was 1.06×10^{-15} , and CFL remained at or below 0.00609. This validates the revised software and I/O cadence, not full-mesh stability.

5.4 Spatial-resolution acceptance

The full-grid pilot must demonstrate:

1. no high-wavenumber energy pile-up in either periodic direction;
2. mean and Reynolds-stress profiles consistent with the MKM database;
3. streamwise and spanwise two-point correlations nearly zero at half-box separation;
4. at least 13 points below $y^+ = 10$;
5. symmetry of the two channel halves;
6. small divergence and no-slip constraint residuals.

6 Two-stage production sampling

6.1 Spin-up

The development stage writes no target snapshots and forces a checkpoint at the end of each restartable segment. It records bulk flux, wall shear, measured Re_τ , all three kinetic-energy components, divergence, CFL, mean profiles, Reynolds stresses, and spectral-tail ratios. After the refined-step $t = 1$ calibration, use five-outer-unit restart segments when the measured wall time fits the allocation. Candidate physical reviews are made at outer times 20, 40, 60, 80, 100, and 120. The user-selected sampling start is fixed at $t = 120$, so no retained temporal sampling begins earlier. The old $t = 20$ spin-up is not automatically transferred to the higher-Reynolds-number case.

A checkpoint is accepted only when bulk flux and measured Re_τ are steady, both walls agree, energy components show no block drift, mean and Reynolds profiles have settled, and spectral/correlation checks pass. A planning calibration criterion is

$$\frac{|Re_{\tau, \text{meas}} - 395|}{395} \leq 0.02. \quad (17)$$

6.2 Stationary sampling

The sampling stage restarts from the accepted $t = 120$ checkpoint into a separate velocity file and ends at $t = 300$. Thus

$$t_{\text{sample,start}} = 120, \quad t_{\text{sample,end}} = 300, \quad T_{\text{sample}} = 180. \quad (18)$$

At the refined step these endpoints are DNS steps 2,400,000 and 6,000,000, respectively, so the stationary stage contains 3,600,000 DNS steps. The dense cadence used for temporal observables is

$$\boxed{\Delta t_{\text{temporal}} u_\tau / h = 0.005}. \quad (19)$$

For Eq. (16), this is one temporal sample every 100 DNS steps and

$$\Delta t_{\text{temporal}}^+ = Re_\tau \Delta t_{\text{temporal}} = 1.975. \quad (20)$$

The corresponding temporal Nyquist angular frequency is

$$\omega_{\text{Ny}} h / u_\tau = \frac{\pi}{\Delta t_{\text{temporal}}} = 628.319. \quad (21)$$

Dense full fields are not the primary temporal record. At the dense cadence the writer stores the complete independent MKM spectral state, from which all three velocity components are reconstructed on every one of the 192 wall-normal planes during postprocessing. All-wall mean and Reynolds-stress raw moments are also accumulated by time block as an independent audit. A complete physical audit velocity field is retained only every 10 outer time units, or every 200000 DNS steps, and at the final accepted state. State data are written in 1000-sample shards (five outer time units), with a forced checkpoint every five outer units. Convergence is reviewed over accumulated 40-outer-time-unit blocks.

A short pilot records temporal observables every 50 DNS steps ($\Delta t = 0.0025$, $\Delta t^+ = 0.9875$) and compares them with the same record decimated to 0.005. The production cadence is reduced to 0.0025 only if the 0.005 correlations or spectra show meaningful aliasing in the frequency band of interest.

For the fixed 180-unit stationary window, Eq. (19) gives 36,000 samples. The all-plane temporal auto-correlations and auto-spectra are evaluated from reconstructed float64 velocity batches in the same time order used by the existing estimators. A Hann/Welch estimate with 8,192-sample segments and 50 percent overlap has segment duration 40.96 and angular-frequency spacing about 0.1534. A full-record diagnostic periodogram has spacing about 0.03491.

The datasets `lag_covariance/lag_0` and `lag_covariance/lag_1` are not required and will not be generated. Any legacy postprocessor invocation uses `--max-lag -1`.

Candidate retained windows are evaluated with at least 12 blocks. The first field-statistics gate is the established five-percent tolerance for both mean and Reynolds-stress profiles. A three-percent sensitivity check is preferred before final acceptance. Scalar energy drift is reported separately rather than hidden by the field-profile result.

6.3 Exact independent-state record

The KMM implementation advances the wall-normal velocity coefficient field $\hat{u}_y$ in its biharmonic space and the wall-normal vorticity $\hat{\eta}_y$ in its Dirichlet space. For a nonzero horizontal wavenumber pair (κ, λ) , the same reconstruction used by `demo/ChannelFlow.py` is

$$\hat{u}_x = i \frac{\kappa D_y \hat{u}_y + \lambda \hat{\eta}_y}{\kappa^2 + \lambda^2}, \quad (22)$$

$$\hat{u}_z = i \frac{\lambda D_y \hat{u}_y - \kappa \hat{\eta}_y}{\kappa^2 + \lambda^2}. \quad (23)$$

The two $(\kappa, \lambda) = (0, 0)$ profiles, $\hat{u}_x(y; 0, 0)$ and $\hat{u}_z(y; 0, 0)$, are evolved by separate one-dimensional equations and must be saved explicitly. Therefore each temporal sample contains

$$\{\hat{u}_y, \hat{\eta}_y, \hat{u}_x(\cdot; 0, 0), \hat{u}_z(\cdot; 0, 0)\}.$$

Restoring these objects and applying the solver's own derivative, wavenumber, zero-mode insertion, and backward-transform paths recovers every velocity component at every saved wall-normal point. The state record uses complex128/float64 and the native real-to-complex Fourier half-spectrum. Wall-normal subsampling, horizontal mode truncation, reduced precision, and lossy compression are excluded from the baseline.

7 Sampled target objects

For retained samples t_n , the horizontal/time mean is

$$\bar{u}_\alpha(y_m) = \frac{1}{N_s N_x N_z} \sum_{n,j,\ell} u_\alpha(x_j, z_\ell, y_m, t_n). \quad (24)$$

With $u'_\alpha = u_\alpha - \bar{u}_\alpha$, the Reynolds profile is

$$R_{\alpha\beta}(y_m) = \frac{1}{N_s N_x N_z} \sum_{n,j,\ell} u'_\alpha(x_j, z_\ell, y_m, t_n) u'_\beta(x_j, z_\ell, y_m, t_n). \quad (25)$$

The normalized horizontal Fourier coefficients are

$$\hat{u}_\alpha(y_m; \kappa_p, \lambda_q, t_n) = \frac{1}{N_x N_z} \sum_{j,\ell} u'_\alpha(x_j, z_\ell, y_m, t_n) e^{-i(\kappa_p x_j + \lambda_q z_\ell)}. \quad (26)$$

For one mode the target stacks the three components level by level into

$$\hat{\mathbf{q}}_{pq} = [\hat{u}_x(y_0), \hat{u}_z(y_0), \hat{u}_y(y_0), \dots, \hat{u}_x(y_{191}), \hat{u}_z(y_{191}), \hat{u}_y(y_{191})]^T \in \mathbb{C}^{576}. \quad (27)$$

The equal-time covariance is

$$B_{pq}^{(0)} = \frac{1}{N_s} \sum_n \hat{\mathbf{q}}_{pq}(t_n) \hat{\mathbf{q}}_{pq}(t_n)^*. \quad (28)$$

8 Constraint on the even Chebyshev–Gauss grid

For each horizontal mode, define component selectors

$$E_x \hat{\mathbf{q}} = \hat{\mathbf{u}}_x, \quad E_y \hat{\mathbf{q}} = \hat{\mathbf{u}}_y, \quad E_z \hat{\mathbf{q}} = \hat{\mathbf{u}}_z,$$

where here E_y selects wall-normal velocity and E_z selects spanwise velocity in physical notation. In the target implementation the selector names must follow the stored component metadata rather than ambiguous coordinate letters.

Let D_y be the barycentric nodal derivative on the 192 Gauss points and let $B_\partial \in \mathbb{R}^{2 \times 192}$ interpolate those points to $y = \pm 1$. The modal divergence and boundary rows are

$$G_{\text{div}}(\kappa, \lambda) = i\kappa E_{\text{stream}} + i\lambda E_{\text{span}} + D_y E_{\text{wall}}, \quad (29)$$

$$G_\partial = \begin{bmatrix} B_\partial E_{\text{stream}} \\ B_\partial E_{\text{span}} \\ B_\partial E_{\text{wall}} \end{bmatrix}. \quad (30)$$

Thus

$$G_{\text{raw}}(\kappa, \lambda) = \begin{bmatrix} G_{\text{div}} \\ G_{\partial} \end{bmatrix} \in \mathbb{C}^{198 \times 576}. \quad (31)$$

If

$$G_{\text{raw}} = U \Sigma V^*,$$

and V_r contains the right singular vectors retained at the documented relative tolerance, the saved row-orthonormal constraint is

$$\tilde{G} = V_r^*, \quad \tilde{G} \tilde{G}^* = I, \quad qquad \tilde{G} \hat{q} = 0. \quad (32)$$

The expected generic row count is 198, with two dependencies commonly present at the zero horizontal mode; the actual rank table must be computed and saved, not assumed.

9 Storage scaling and processing decision

The proposed mesh has 49,152 horizontal modes and modal dimension 576. The uncompressed estimates in Table 2 show that the old output policy cannot be copied blindly.

Table 2: Approximate uncompressed storage for the proposed mesh.

Object	Size
One three-component physical snapshot	0.210938 GiB
36,000 physical velocity samples	7.4158 TiB
One independent spectral-state sample	0.142093 GiB
36,000 independent state samples	4.9954 TiB
18 complete audit fields	3.797 GiB
Full equal-time <code>modal/B0_DNS</code>	approximately 243 GiB

For the real-to-complex horizontal layout, the nonredundant coefficient extent is $192 \times 256 \times 97$. Two complex128 state arrays and two float64 zero-mode profiles contain 152,570,880 bytes per sample. This exact representation saves approximately 32.64 percent relative to three dense physical components while retaining every horizontal plane. State files are split into 1000-sample shards, each approximately 142.09 GiB before compression.

Only representation-preserving reductions are admitted: the native Fourier half-spectrum, the independent state variables, and benchmarked lossless compression. Compression ratios are not assumed in capacity planning. Float32, lossy filters, mode truncation, wall-normal subsampling, and further temporal decimation are excluded unless a separate error study is accepted. The server allocation should be at least 7 TiB for the baseline state, checkpoints, audit fields, derived outputs, and safety headroom, excluding any large disposable reconstruction cache. A full B_0 requires an explicit scientific need and an additional I/O benchmark. Lag covariances are omitted unconditionally.

The detailed sufficient-data construction and its numerical-equivalence test are recorded in `scripts/mkm_dns_target/TEMPORAL_SAMPLING_REMEDY_Re_tau_395.md`.

Before production, at least 512 samples are dual-written as conventional physical fields and as independent state. Reconstructed coefficients, physical velocities, mean/Reynolds profiles, and temporal correlations and spectra at all 192 planes must agree to bitwise identity where possible and otherwise to a documented roundoff/FFT-order tolerance. The test must include the zero horizontal mode, two MPI decompositions, shard interruption recovery, lossless decompression checksums, and measured write throughput. The existing constraint builder and postprocessors must also be benchmarked and, if necessary, optimized at 49,152 horizontal modes.

10 Planned command card

After the preflight gates have passed, the common run settings are intended to resolve as follows:

```
TMPDIR=/media/jay/data1/tmp
ENV=/media/jay/data1/conda_envs/shenfun_dns_np126_20260702
REPO=/media/jay/data1/shenfun
OUT=/media/jay/data1/shenfun_dns_runs/
    production_mkm_Re395_N192_256_192
PREFIX=MKM_production_Re395_N192_256_192

--n 192 256 192
--domain -1 1 0 6.283185307179586 0 3.141592653589793
--re 395
--bulk-velocity 17.54475114359453
--dt 0.00005
--padding-factor 1.5 1.5 1.5
--family C
--timestepper IMEXRK222
--save-mesh quadrature
```

The 64-rank candidate reserves 128 SLURM tasks. The refined time step remains subject to the $t = 1$ acceptance pilot. The selected spin-up end and retained sampling window are fixed at $t = 120$ and $120 \leq t \leq 300$; the scripts reject checkpoints or segment endpoints outside that schedule.

11 Acceptance record

The final handoff will include the code revision, resolved initialization, integer padded sizes, server environment, MPI benchmark, selected Δt , measured CFL, time-averaged wall stresses and actual Re_τ , accepted spin-up checkpoint, sampling-window convergence, reference-profile and spectral comparisons, half-box correlations, constraint ranks and residuals, file sizes/checksums, the exact dense temporal cadence and estimator settings, the state-reconstruction version and round-trip report, and a list of deliberately omitted large products. It will explicitly record that `lag_0` and `lag_1` were not generated.

12 Conclusion

The reference paper’s odd count is a Lobatto-grid convention, but the current optimized Shenfun MKM biharmonic solver requires an even wall-normal size. The selected $N_{\text{wall}} = 192$ preserves the reference center resolution closely and is vastly finer than the old 64-point mesh, while Shenfun continues to impose the walls through its Galerkin bases. The decisive remaining prerequisites are an $Re_\tau = 395$ -aware initialization and flux target, explicit upward rounding of the de-aliasing sizes, measured wall-stress diagnostics, the refined full-shape time-step pilot, and a storage-aware postprocessing policy. Long production should begin only after the refined-step and physical-acceptance gates have passed.